

Heart Disease Prediction based on AI

Human-AI Cooperation: Human in the Loop, Explainability, Trustworthiness and Transparency in AI

Benjamim Moreira

José Filipe

José Cunha

Universidade de Coimbra

uc2020261856@student.uc.pt, josepedrofilipe1@gmail.com, josecunha.8989@gmail.com

Abstract

This project focuses on leveraging machine learning techniques, including passive and active learning, to assist in the early detection of heart disease based on various predictive features. By employing evaluation metrics, the study aims to identify the most effective models to support medical experts in diagnosing heart conditions. Among the models tested, CatBoost demonstrated exceptional performance in the active learning framework. The study explores multiple strategies for sample selection, such as random sampling, entropy sampling, least confident sampling, and margin sampling, to optimize model accuracy and efficiency. Additionally, interpretability tools like SHAP and LIME are utilized to provide a comprehensive understanding of feature contributions and model behavior, enhancing trust and usability in clinical decision-making.

1 Introduction

Cardiovascular diseases (CVDs) are the leading global cause of death, accounting for nearly one-third of all fatalities annually. These diseases often progress silently, showing no symptoms until advanced stages, making early detection crucial for saving lives. Early diagnosis can simplify treatment, reduce invasiveness, and improve outcomes.[3] Machine learning[2] offers significant potential in identifying patterns in health data, such as age, cholesterol levels, and blood pressure, which are critical for early detection. However, for AI systems to gain acceptance in healthcare, they must be transparent and interpretable. A "black box" model, no matter how accurate, is unlikely to be trusted if its decision-making process is unclear. Our project addresses this challenge by developing a machine learning system that not only predicts heart disease risk but also explains its reasoning in a way doctors can understand and validate. We integrate SHAP (SHapley Additive Explanations) and LIME (Local Interpretable Model-agnostic Explanations) to clarify the contribution of

individual factors, like cholesterol and age, to predictions. To enhance performance with limited data, we employ active learning strategies—such as random sampling, entropy sampling, margin sampling, and least confident sampling—alongside robust passive learning models like Random Forest, AdaBoost, Decision Tree, and CatBoost. By combining accuracy with transparency, our system aims to bridge the gap between AI technology and real-world healthcare, enabling earlier and more effective intervention for heart disease.

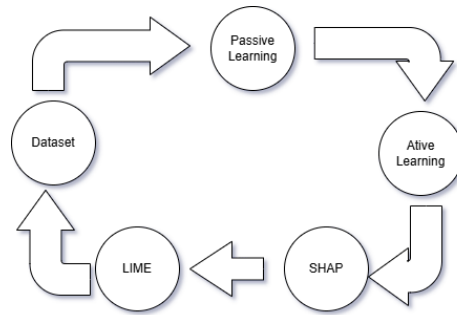


Figure 1: Heart Disease Machine Learning Workflow

The remainder of this paper is organized as follows. Section 2 provides a general review of the state of the art in heart disease prediction and the use of machine learning models, with a focus on interpretability and active learning. Section 3 explains the tools, the dataset utilized in the experimental setup. Sections 4 and 5 detail the methodologies and experimental design employed to achieve the previously defined objectives. Section 4 focuses on explaining each method used in the study, while Section 5 compares the results and discusses the performance of each method in detail, including their respective outcomes. Section 6 concludes the paper.

2 Related Work

Heart disease prediction has been a significant focus in artificial intelligence research, particularly in machine learning, due to its

potential to improve early detection and intervention. Various studies[6] have explored different models and techniques to enhance prediction accuracy and ensure interpretability, addressing the need for reliable diagnostic tools. For example, Chowdhury[11] utilized linear classification models to analyze electrocardiogram (ECG) waves, aiming to diagnose myocardial infarction and prevent accidents. Similarly, Srinivas[10] explored approaches starting with IF-THEN rules and progressing to machine learning techniques such as ID3 Decision Trees, Neural Networks, Stochastic Backpropagation Algorithms, and Bayesian Networks. More recent studies have incorporated ensemble methods and explainability techniques. Samanyu K.[8] applied models like Logistic Regression, Decision Trees, Random Forests, and Support Vector Machines to predict heart disease, focusing on metrics such as accuracy, precision, and recall. However, this work lacked an in-depth exploration of interpretability techniques critical for clinical use. In contrast, Dksonl[1] emphasized both predictive accuracy and explainability by employing CatBoost, a gradient boosting algorithm, alongside SHAP. This approach provided detailed insights into how individual features influenced predictions, achieving a recall of 100 percent, which is crucial for identifying all positive cases in medical diagnostics. This balance of accuracy and transparency highlights the growing importance of interpretability in AI-driven healthcare solutions.

3 Material

The dataset used in this work was created by merging five independent datasets, making it one of the largest consolidated heart disease datasets available for research purposes. The original datasets come from various sources, including clinics and hospitals in Cleveland, Hungary, Switzerland, Long Beach VA, and the Stalog Heart dataset. In total, the original datasets contributed 1,190 observations, with the following distribution: 303 from Cleveland, 294 from Hungary, 123 from Switzerland, 200 from Long Beach VA, and 270 from the Stalog Heart dataset. After harmonizing the data and removing duplicates, the final dataset contains 918 samples and 11 common features. These features include essential medical and demographic attributes such as age, sex, chest pain type, resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise-induced angina, ST depression, and thalassemia status.

Feature	Type	Description
Age	Numerical	Age of the individual (in years)
Sex	Categorical	Biological sex of the individual (e.g., 0 = female, 1 = male)
Chest Pain Type	Categorical	Type of chest pain experienced (e.g., typical angina, atypical)
RestingBP	Numerical	Resting blood pressure (in mm Hg)
Cholesterol	Numerical	Cholesterol level (in mg/dL)
FastingBS	Binary	Fasting blood sugar superior to 120 mg/dL (0 = no, 1 = yes)
RestingECG	Categorical	Resting electrocardiogram results (e.g., normal, ST-T changes)
MaxHR	Numerical	Maximum heart rate achieved during exercise
ExerciseAngina	Binary	Exercise-induced angina (0 = no, 1 = yes)
Oldpeak	Numerical	ST depression induced by exercise relative to rest
ST Slope	Categorical	Slope of the ST segment during peak exercise (e.g., upsloping)
HeartDisease	Binary	Diagnosis of heart disease (0 = no, 1 = yes)

Table 1: Description of features in the heart disease dataset

Previous studies utilizing these features have demonstrated the dataset’s suitability for predicting heart disease, offering robust insights despite its relatively compact feature set.

4 Methods & approach

Passive Learning:

In our passive learning approach for the heart disease project, we implemented four powerful machine learning models: Random Forest, AdaBoost, Decision tree and CatBoost. These models were evaluated using key classification metrics to assess their performance and effectiveness in predicting heart disease. The chosen metrics such as Accuracy, Precision, Recall, and F1-Score,[4] offer a comprehensive understanding of the models’ ability to balance correct predictions and misclassifications.

Metric	Formula	Description
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Proportion of total correct predictions to all predictions.
Precision	$\frac{TP}{TP+FP}$	Proportion of true positive predictions out of all positive predictions made by the model.
Recall	$\frac{TP}{TP+FN}$	Proportion of true positive cases identified out of all actual positives.
F1-Score	$2 \times \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	Harmonic mean of Precision and Recall, balancing their contributions in one metric.

Table 2: Metrics evaluation formula, Precision, Accuracy, F1-Score and Recall

Active Learning:

In the active learning[9] approach, we applied the same set of classifiers used during the passive learning phase, including Decision Tree, Random Forest, AdaBoost, and CatBoost. The primary objective was to iteratively train the models with a small initial labeled dataset, gradually expanding it with

new, informative samples selected through various sampling strategies. Four popular sampling methods were explored: Random Sampling, Entropy Sampling, Margin Sampling, and Least Confident Sampling.

Sampling Method	Description
Random Sampling	Select data points randomly without prioritizing informativeness.
Entropy Sampling	Targeted samples with the highest Shannon entropy in their predicted probability distributions.
Margin Sampling	Emphasized the difference in probabilities between the two most likely classes.
Least Confident Sampling	Selected instances where the model's predicted probability for the most likely class was the lowest.

Table 3: Description of Sampling Methods

These strategies are designed to prioritize uncertain or diverse data points, ensuring that the models focus on challenging instances to improve performance with fewer labeled samples.

The experiments began with 5 % of the labeled dataset and progressively added samples in batches of different sizes (1, 3, 5, and 10), up to a total of 600 labeled samples. To further refine the process, an initial warm-up phase using Random Sampling was implemented, helping to establish a broad and diverse foundation before switching to the active learning strategies. This setup allowed us to investigate how factors such as sampling strategies, batch sizes, and the warm-up phase affected the models' learning curves and overall performance.

SHAP (SHapley Additive Explanations):

Shap[4] will be used as a powerful interpretability tool in machine learning to explain the predictions of complex models by attributing the influence of each feature on a model's output. It is based on cooperative game theory, specifically the Shapley values concept, which ensures a fair distribution of contributions among features.

LIME (Local Interpretable Model agnostic Explanations):

LIME[7] will be used as a tool to make machine learning models more interpretable by providing simple, local explanations for individual predictions. It works by approximating a complex model's behavior with a simpler, interpretable model (such as a linear regression) around the neighborhood of the specific instance being explained.

5 Experiments

5.1. Passive Learning vs Active Learning

5.1.1 Passive Learning:

Passive Learning utilizes the entire labeled dataset without implementing iterative sample selection strategies. The models evaluated in this context were the Decision Tree Classifier, Random Forest Classifier, AdaBoost Classifier, and CatBoost Classifier[5]. These models were chosen for their diversity in methodology and effectiveness in classification tasks proven in previous works.

The results, obtained using a training size of 75% and a test size of 25%, are presented for both the training and test datasets to assess the models' learning capabilities and generalization performance. Evaluation of classification reports, detailing metrics such as precision, recall, F1-score, and support, providing insights into the models' performance across different classes. Training results:

	Classification Algorithms	Training Accuracy	Training Precision	Training Recall	Training F1 Score
0	Decision Tree Classifier	0.867733	0.847059	0.932642	0.887793
1	Random Forest Classifier	0.878453	0.859189	0.932642	0.894410
2	AdaBoostClassifier	0.876453	0.886889	0.893782	0.890323
3	CatBoostClassifier	0.917151	0.906173	0.950777	0.927939

Figure 2: Passive learn training results

In terms of training accuracy, the CatBoost Classifier outperformed the other models with 91.7%, followed by Random Forest and AdaBoost (both 87.6%) and Decision Tree (86.7%). For the F1-Score, which balances precision and recall, CatBoost again led with 92.7%, with Random Forest (89.4%), AdaBoost (89.0%), and Decision Tree (88.7%) trailing behind.

When evaluating precision and recall, CatBoost excelled, achieving a precision of 90.6% and recall of 95.1%, reflecting its strong performance in correctly identifying positive classes and minimizing false negatives. AdaBoost had the highest precision among the other models at 88.7%, but its recall was lower than CatBoost's.

Overall, CatBoost demonstrated the best performance among the classifiers in the testing results.

	Classification Algorithms	Testing Accuracy	Testing Precision	Testing Recall	Testing F1 Score
0	Decision Tree Classifier	0.800000	0.763889	0.901639	0.827068
1	Random Forest Classifier	0.865217	0.832117	0.934426	0.880309
2	AdaBoostClassifier	0.865217	0.858268	0.893443	0.875502
3	CatBoostClassifier	0.860870	0.835821	0.918033	0.875000

Figure 3: Passive learn testing results

In the test results, Random Forest and AdaBoost achieved the highest accuracy at 86.5%, slightly outperforming CatBoost at 86.1%. The Decision Tree Classifier had the lowest accuracy at 80.0%. In terms of F1-Score, Random Forest scored 87.5%, while AdaBoost was close with 88%.

Regarding recall, Random Forest led with 93.4%, outperforming the other models.

However, its precision (83.2%) was slightly lower than CatBoost's (83.5%). AdaBoost achieved the highest precision at 85.8% and its recall was slightly lower than CatBoost's (91.0%). These results highlight CatBoost's strong performance in training and its balanced performance in testing, positioning it as the most reliable model for heart disease prediction. and accuracy, particularly in healthcare applications.

5.1.2 Active Learning:

5.1.2.1 Active Learning Strategies and Sampling Approaches:

In this study, active learning techniques were applied to the same set of classifiers used in the passive learning phase: Decision Tree, Random Forest, AdaBoost, and CatBoost. Various sampling strategies were explored, including Random Sampling, Entropy Sampling, Margin Sampling, and Least Confident Sampling, to assess their impact on model performance. These strategies were designed to guide sample selection during the iterative training process, focusing on uncertain or diverse data points to improve the classifiers' generalization ability.

The experiments involved all four classifiers across each sampling strategy, with batch sizes of 1, 3, 5, and 10, starting with 5% of the dataset and expanding up to 600 samples (65.35% of the total dataset). This incremental approach enabled the evaluation of how the number of samples selected in each iteration influenced model performance, and how different sampling techniques affected the active learning process.

Decision tree

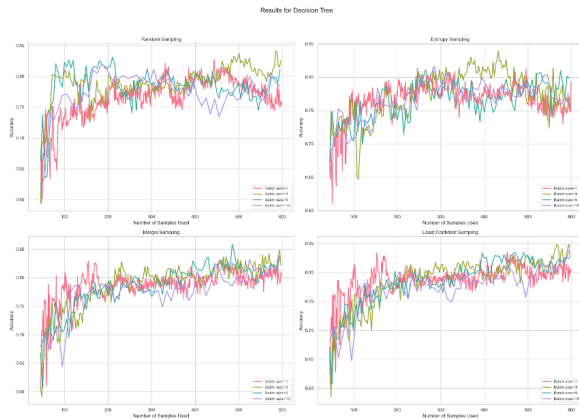


Figure 4: Decision Tree results for different batch sizes of sampling methods

The Decision Tree model proved to be unstable, showing rapid evolution across all sampling strategies but with significant variability, never stabilizing its accuracy. Regarding batch sizes, smaller batches led to slightly faster progress, especially with uncertainty-based strategies.

Random forest

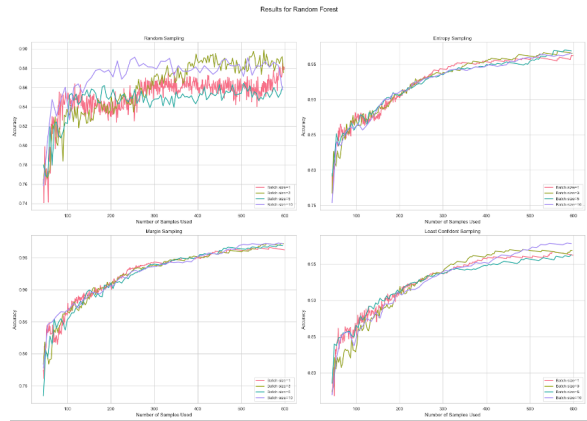


Figure 5: Random Forest results for different batch sizes of sampling methods results

Random Forests displayed a notable advantage when using uncertainty-based strategies, achieving significant accuracy gains in the early stages of training and having stabilised the accuracy. Larger batches resulted in smoother trends but slightly reduced early-stage effectiveness. As for random sampling, it has a large deviation in accuracy even when the other strategies have already stabilised their accuracy

AdaBoost

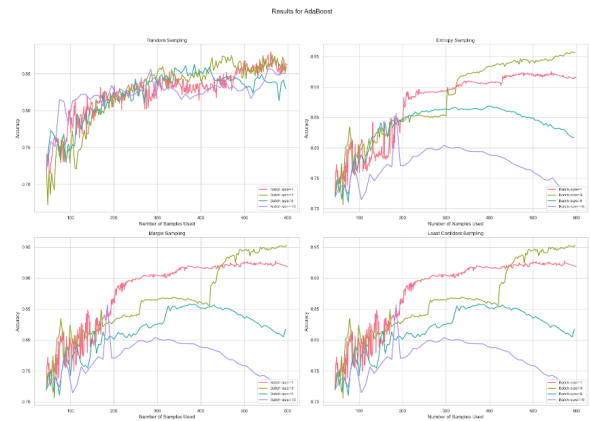


Figure 6: AdaBoost results for different batch sizes of sampling methods

AdaBoost is a particular case in which we see a lot of contrast in the different strategies. Uncertainty-based strategies similarly outperformed Random Sampling for smaller batch size but for larger batch sizes it couldn't stabilise its accuracy, the simultaneous introduction of a large number of samples introduced confusion to the model. With regard to random sampling, we continue to see a large deviation in the results

Catboost

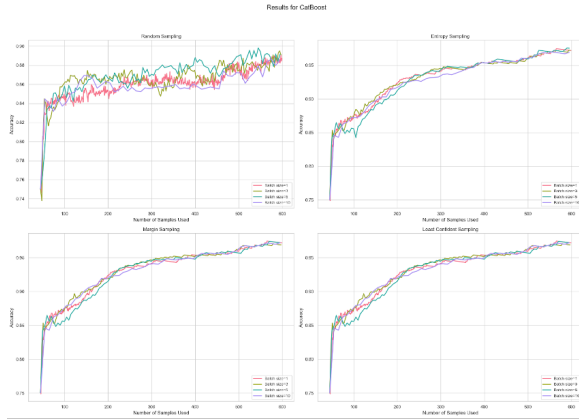


Figure 7: CatBoost results for different batch sizes of sampling methods

CatBoost also demonstrated strong performance with uncertainty-based strategies. Achieving significant accuracy improvements early in the process. Smaller batch sizes also enabled more focused updates, leading to faster learning. Random Sampling, while steadily improving accuracy, was consistently less efficient compared to these targeted approaches.

Boxplot of the results obtained:

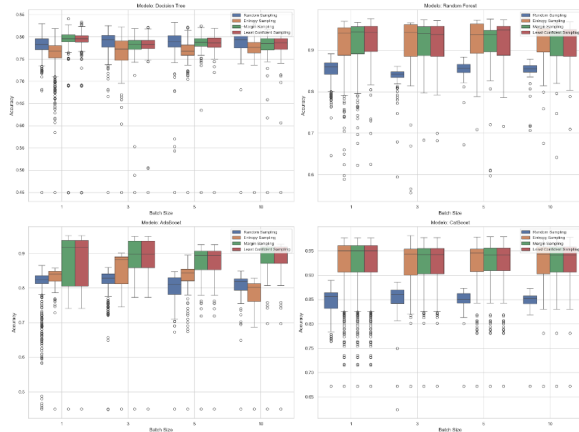


Figure 8: boxplot for models and batches

Analysing the accuracy boxplot for models and batch we verify that CatBoost and Random Forest demonstrate strong performance, maintaining high accuracy regardless of the sampling strategy or batch size. Decision Tree struggles overall, with accuracy remaining low and relatively unaffected by the sampling strategy. Adaboost shows differences in the results, with entropy sampling not providing consistency for the different batch sizes. Smaller batch sizes (1–3) generally yield more variability in all models, while larger batches (5–10) stabilize accuracy. Overall, CatBoost emerges as the most reliable model, while AdaBoost the one with more divergence in results

Accuracy boxplots for models and batch Sizes:

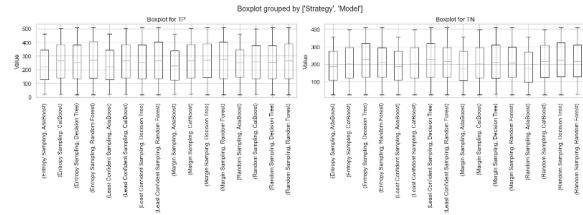


Figure 9: Boxplot for models and batches: TP vs TN

The Accuracy boxplots for models and batch Sizes graphs evaluate the confusion matrix components (TP, TN, FP, FN) across models and sampling strategies. The results show that differences in these metrics are more influenced by the choice of the model than the sampling strategy. Decision tree and Random Forest consistently achieve high TP and TN values, reflecting their superior classification ability.

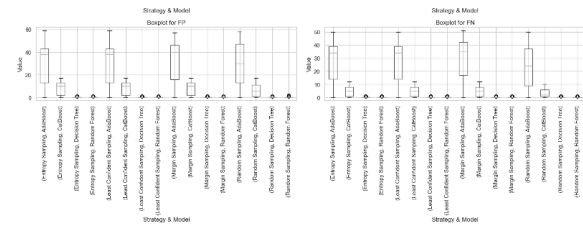


Figure 10: Boxplot for models and batches: FP vs FN

AdaBoost shows higher variability in FP and FN, indicating less robust performance. Catboost has balanced results but occasionally higher FN rates, suggesting potential challenges in identifying minority cases. Sampling strategies seem to have minimal impact compared to the intrinsic model capabilities.

5.1.2.2 Study on how the introduction of warm-up samples (selected with random sampling) affects results

This section investigates how introducing warm-up samples selected through random sampling influences active learning performance. The experiments apply active learning to four models, using different batch sizes and sampling strategies (Entropy, Margin, Least Confident). Starting with 5% of the dataset, random sampling is used until 300 samples are reached, after which the specified strategies are implemented to further optimize model performance.

Decision tree



Figure 11: Decision Tree results for different batch sizes and sampling methods with 300 warm up samples

The Decision Tree model performed similarly to the tests carried out without the introduction of warm up samples, while continuing to show difficulties in stabilising its accuracy.

Random Forest

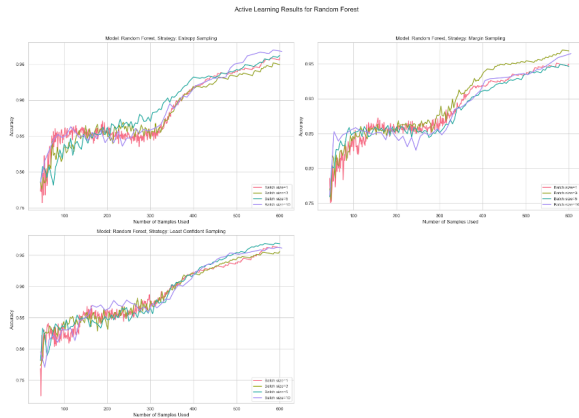


Figure 12: Random Forest results for different batch sizes and sampling methods with 300 warm up samples

The performance of the Random Forest model aligns with expectations, as it typically benefits from a larger number of samples to refine its predictions and achieve optimal accuracy. This characteristic stems from its reliance on aggregating results from multiple decision trees, which perform better with more diverse and representative data.

AdaBoost

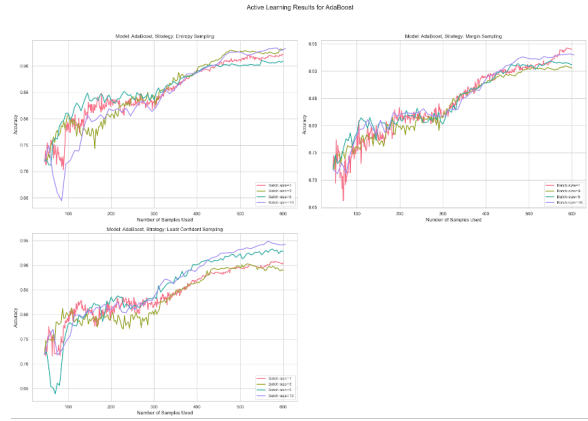


Figure 13: AdaBoost results for different batch sizes and sampling methods with 300 warm up samples

Looking at the performance of AdaBoost, it is possible to see that the introduction of warm-up samples for the construction of the model allowed it to stabilize its accuracy, which did not happen when active learning was applied without the warm-up samples. The model benefits significantly from the initial 300 random samples before transitioning to more targeted sampling strategies.

CatBoost:

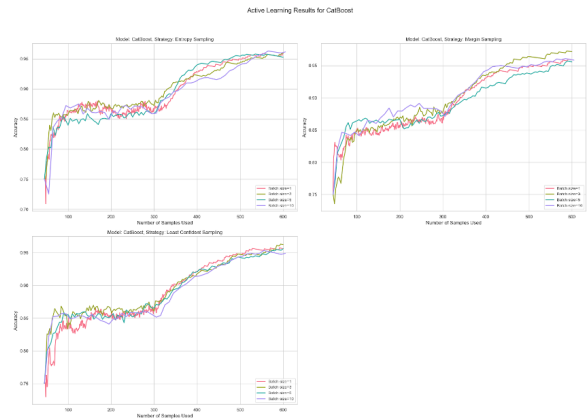


Figure 14: CatBoost results for different batch sizes and sampling methods with 300 warm up samples

For CatBoost, adding 300 warm-up samples slowed accuracy convergence, proving ineffective. In contrast, using random selection until 300 samples helped stabilize precision in AdaBoost. However, for CatBoost and Random Forest, this approach delayed accuracy improvement, while for the Decision Tree, it made little difference compared to the no-warm-up approach.

5.2 Interpretability techniques

5.2.1 SHAP (SHapley Additive exPlanations): It was developed a SHAP model to identify and visualize the most significant features influencing heart disease predictions.

5.2.1.1 Shap Patients with heart disease

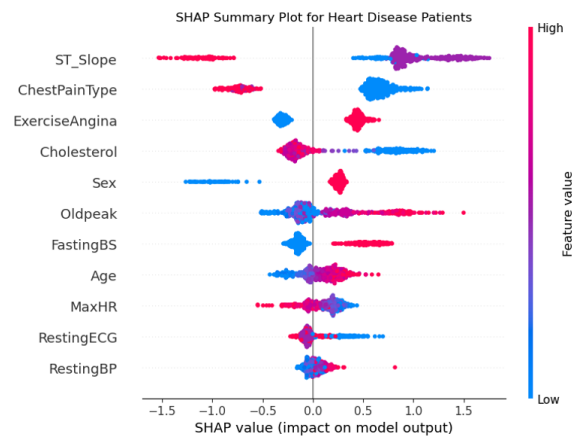


Figure 15: Shap values: Patients with heart disease

In patients with heart disease, features such as STSlope, Oldpeak, ChestPainType, ExerciseAngina, and Cholesterol emerge as strong predictors. The notable SHAP values for these features suggest that they are critical indicators in identifying heart disease, with higher values correlating with a greater risk. This insight emphasizes the need to focus on these factors in both clinical assessments and preventive measures.

5.2.1.2 Shap Patients without heart disease

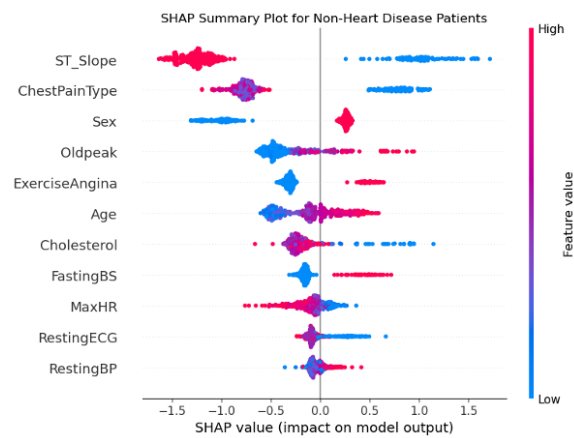


Figure 16: Shap values: Patients without heart disease

In the non-heart disease group, MaxHR and Age become more significant, while STSlope and Oldpeak remain important. Higher MaxHR and older age correlate with non-heart disease predictions, suggesting these factors may indicate better heart function. The color gradient and SHAP analysis show that high STSlope and Oldpeak values boost heart disease predictions, with a higher Sex value (representing males) also linked to heart disease, though less so in the non-heart disease group. Heart

disease patients have a wider SHAP value range, indicating stronger feature influence compared to the more clustered values in the non-heart disease group.

5.2.2 LIME (Local Interpretable Model-agnostic Explanations)

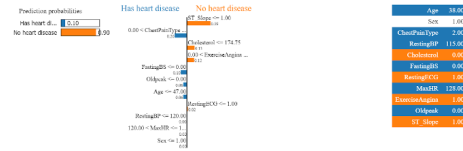


Figure 17: LIME Visualization of Model Prediction for Heart Disease: Feature Contributions and Input Data

The model's high confidence in predicting no heart disease comes from features that suggest a lower risk, such as an upsloping ST segment (STSlope), low cholesterol levels, and the absence of exercise-induced angina (ExerciseAngina = 0). These factors align with a healthy heart, prompting the model to predict no heart disease.

However, ExerciseAngina, indicating chest pain during exercise, remains a notable feature. While it has little impact on the current prediction, it could warrant further clinical investigation, as chest pain may sometimes signal early heart issues.

6 Conclusions

This study identified effective strategies to optimize machine learning models for heart disease prediction, with Active Learning proving more robust than Passive Learning, particularly in uncertainty-based sample selection. Using just 65.35% of the dataset, Active Learning outperformed Passive Learning in accuracy, with smaller batch sizes enabling quicker performance stabilization. Notably, introducing warm-up samples enhanced stability in the AdaBoost model.

Interpretability was a key focus, with SHAP uncovering critical features like ST.Slope, ChestPainType, and ExerciseAngina. LIME further demonstrated how upsloping ST segments and the absence of exercise-induced angina were linked to lower heart disease risk, bridging machine learning predictions with clinical insights.

These findings advance human-centered AI by improving both model performance and interpretability. Future work could explore larger datasets, feature selection, and new models to further enhance predictive accuracy and reliability in healthcare applications.

Note on the Reduced Version

This paper has been edited to comply with the page limits required by the submission format. The full version, including expanded sections, additional analyses, and supplementary materials, is available at: <https://drive.google.com/file/d/1iYw8wsI6uc-U068xEUOtO9MiLy1QhJjS/view?usp=sharing>

References

- [1] DKson1. *Heart Disease Explainable CatBoost 100% Recall*. Kaggle Notebook. Accessed: 2024-11-29. 2024. URL: <https://www.kaggle.com/code/dkson1/heart-disease-explainable-catboost-100-recall/notebook>.
- [2] Z. Obermeyer and E. J. Emanuel. “Predicting the future — Big data, machine learning, and clinical medicine”. In: *New England Journal of Medicine* 375.13 (2016), pp. 1216–1219. DOI: 10.1056/NEJMp1606181.
- [3] World Health Organization. “Cardiovascular diseases (CVDs) fact sheet”. In: (2019). Accessed: 2024-10-08. URL: [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)).
- [4] D. M. Powers. “Evaluation: From Precision, Recall and F-Factor to ROC, Informedness, Markedness & Correlation”. In: *Journal of Machine Learning Technologies* 2.1 (2011), pp. 37–63. URL: <https://arxiv.org/abs/2010.16061>.
- [5] L. Prokhorenkova et al. “CatBoost: unbiased boosting with categorical features”. In: *Advances in Neural Information Processing Systems*. 2018, pp. 6639–6649. URL: https://papers.nips.cc/paper_files/paper/2018/file/14491b756b3a51daac41c24863285549-Paper.pdf.
- [6] M. T. Ribeiro, S. Singh, and C. Guestrin. ““Why should I trust you?”: Explaining the predictions of any classifier”. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2016, pp. 1135–1144. DOI: 10.1145/2939672.2939778.
- [7] M. T. Ribeiro, S. Singh, and C. Guestrin. ““Why should I trust you?”: Explaining the predictions of any classifier”. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2016, pp. 1135–1144. DOI: 10.1145/2939672.2939778.
- [8] S. Samanyuk. *Heart Disease Prediction*. Accessed: 2024-11-29. 2024. URL: <https://www.kaggle.com/code/samanyuk/heart-disease-prediction/notebook>.
- [9] B. Settles. *Active Learning Literature Survey*. Tech. rep. University of Wisconsin-Madison, 2009. URL: <http://burrsettles.com/pub/settles.activelearning.pdf>.
- [10] K. Srinivas, B. K. Rani, and A. Govrdhan. “Applications of Data Mining Techniques in Healthcare and Prediction of Heart Attacks”. In: *International Journal on Computer Science and Engineering (IJCSE)* 2.02 (2010), pp. 250–255.
- [11] A. Xu et al. “A Wearable System for Daily Activity Monitoring Based on Hidden Markov Model and Intelligent Glasses”. In: *Sensors* 19.12 (2019), p. 2780. DOI: 10.3390/s19122780. URL: <https://www.mdpi.com/1424-8220/19/12/2780>.